

HW8 - Engineering Management

Student Hub Platform

Negative Patterns (and How We Avoid Them)

1. Hiring Pushovers

Problem:

Members of a small student team can tend to take the easiest route and just accept the opinions of others, which results in weak decisions, unnoticed problems, and loss of insightful debate.

How We Implement It:

- Team members are encouraged to provide their opinions.
- Major decisions require input from all the team members.
- Members share their work so that others can review it.

Result:

Better decision-making, stronger technical decisions and improved project quality.

2.Ignore Human Issues

Problem:

Overlooking human issues like stress, university and work obligations, and poor communication can significantly lower team morale and dynamics.

How We Implement It:

- Hold weekly meetings to discuss progress and problems.
- Tasks ought to be divided based on each individual's competencies.
- Use of shared tools (Discord, Trello, GitHub) to maintain communication.

Result:

Better communication, improved teamwork and stable progress.

3. Ignore Low Performers

Problem:

If a team member continuously contributes less and is late with the deadlines, other members are going to be overloaded, and project progress will be significantly slower.

How We Implement It:

- Instead of reassigning tasks, offer guidance and support.
- Provide positive feedback upon milestone completion.
- Collaboration on difficult tasks allows weaker members to gain knowledge.

Result:

Improved teamwork and an environment that promotes development.

Positive Patterns (and How We Implement Them)

1. Lose the Ego

Why It Matters:

Maintaining that one person is always right can lead to conflict and hinder teamwork. Everyone should be receptive to criticism and new ideas.

How We Implement It:

- Mistakes are to be treated as learning opportunities.
- Core decisions are discussed as a group rather than a single person.
- Acknowledge when a different solution is better.

Result:

Better team dynamics and improved decision-making.

Be a Teacher and a Mentor

Why It Matters:

Not everyone has the same experiences and expertise with the required technology. Sharing knowledge helps the team improve and produce a better end product.

How We Implement It:

- Code reviews are used to teach better techniques in addition to identifying errors.
- Help each other understand concepts and technologies
- Instead of criticizing errors, we promote asking questions and learning together.

Result:

Improved team skills and teamwork, and more consistent project quality.